

# Singh Sanskar Surendrasingh

DSIRe Lab, Block-3, IIT Delhi

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sanskar2705

## Research Interests

Information retrieval and multimodal representation learning, with a diagnostic bent: explaining and predicting system behaviour from the geometry of learned representations. Increasingly drawn to NLP and mechanistic interpretability, which ask the same of model internals.

## Education

### Indian Institute of Technology (IIT) Kharagpur

CGPA: 8.46 / 10

Master of Technology in Wireless Communication & Networks

Jun 2023 – May 2025

- Master's Thesis: *Facial Recognition and Verification using a Vision-Transformer Siamese Network* (advised by Prof. Saumik Bhattacharya & Prof. Suva Sekhar Das).

### Gujarat Technological University

CGPA: 7.83 / 10

Bachelor of Engineering in Electrical Engineering

Jun 2019 – May 2023

## Publications

- Scalability Analysis of Cross-Modal Text-to-Image Pipelines.**  
Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath.  
CIKM 2026 – Full Research Paper (*Accepted*).
- Retrieval-Stage Interactions in Cross-Modal Text-to-Image Retrieval Pipelines.**  
Sanskar S. Singh, Ziv Barretto, Bharati Khanijo, Aditya Sahu, Maya Ramanath.  
CIKM 2026 – Short Paper (*Accepted*).
- EduExplain: A Benchmark for Multimodal Explanation Set Retrieval from Educational Resources.**  
Dhruv Tiwari, Sanskar S. Singh, Ritika Jain, Shambhavi Bisht, Maya Ramanath.  
SIGIR-AP 2026 (*under review*).
- SARCH: Multimodal Search for Archaeological Archives.**  
N. Sinha, B. Khanijo, Sanskar S. Singh (3rd author), *et al.*  
CIKM'25 MMGenSR Workshop, Seoul – *Published*.

## Research Experience

### IIT Delhi · Advisor: Prof. Maya Ramanath

New Delhi, India

Project Scientist – Multimodal Information Retrieval, DSIRe Lab

May 2025 – Present

- Cross-modal retrieval pipelines at scale (first author; full paper accepted, CIKM 2026).** Systems-level study of single-index, retrieve-and-rerank, and ensemble pipelines on MS-COCO, Flickr30K, and a 7M-image LAION subset, measuring accuracy, latency, energy, storage, and candidate fidelity. Identified **O-IVFPQ as the strongest single-index trade-off**, with reranking dominating end-to-end cost once applied, making index choice secondary.
- Retrieval-stage interactions in cross-modal retrieval (first author; short paper accepted, CIKM 2026).** Studied how candidate generation, reranking, and fusion interact across retrieval pipelines on MS-COCO and Flickr30K. Found that **reranking gains depend on first-stage retriever quality**, candidate-set size helps reranking but hurts RRF ensembles, and ensemble gains stem from complementary query coverage, not raw accuracy.
- Spatial fidelity maps – query-time prediction of ANN retrieval quality** Candidate fidelity can only be measured by running the exact-KNN search an approximate index exists to avoid; proposed an **offline-built landmark lookup** that predicts per-query CF@100 from a query's nearest landmark alone, with no exact reference at query time. On a **1M-pair LAION pilot** (50K held-out queries; 9 HNSW/IVFPQ/O-IVFPQ configurations), it beat both a global-average and a density-only predictor in **all 16 model-modality-direction pairings** (4 embedding models × 4 query/index directions), with skill **0.17–0.67** (0 = no better than baseline) and mean calibration error  $\leq 0.009$ .
- EduExplain (co-author; under review, SIGIR-AP 2026).** Built the first benchmark for **multimodal explanation-set retrieval**, pairing 321 learner queries with human-validated gold sets across 16 science textbooks; found that diversity-based set selection underperforms simple top-ranked retrieval.
- MMIR-Explorer:** designed and deployed an interactive platform for query-level analysis of retrieval-stage interactions – side-by-side pipeline comparison, reranking effects, candidate provenance, and ANN fidelity. Live at [mmir.iitd.ac.in](http://mmir.iitd.ac.in).
- SARCH – multimodal archival search (published, CIKM'25 Workshop):** co-developed a hybrid retrieval system over 296 scanned archaeology documents (63K+ pages) using modality-specific embeddings (MiniLM, CLIP, TAPAS) fused with BM25 via reciprocal-rank fusion.

## Projects

### IIT Kharagpur

One-Shot Facial Verification with a Siamese Network (Master's Thesis)

Jun 2024 – May 2025

- Trained a **Vision-Transformer Siamese model** on Celeb-DF v2 (31K+ images) for one-shot recognition and verification, using balanced (anchor, positive, negative) triplets, a metric-learning setup that mirrors learned similarity in dense retrieval.
- Achieved **94.83% recognition accuracy** and **91.06% verification accuracy** (98.44% precision, 85.77% recall).

## Industry Experience

### Eternal Robotics

Ahmedabad, India

Computer Vision Intern

Feb 2025 – May 2025

- Integrated and deployed a YOLO-based quality-inspection system for car assembly lines, and built an OCR proof-of-concept on vehicle stickers with Qwen2.5-VL-Instruct (**94.74% accuracy**); contributed to a CI/CD pipeline for model deployment.

### Unizoy

Surat, India

Machine Learning Engineering Intern

May 2024 – Jul 2024

- Developed a real-time pose-estimation and feedback system (Mediapipe) covering 10+ exercises for remote physiotherapy, reaching **93% accuracy** on the biceps-curl model.

### Sard AI

Remote

Machine Learning Intern

May 2023 – Jul 2023

- Added emotion-detection (EfficientNet-B7, **74%** validation accuracy) and gender-classification (VGG16, **93%** on CelebA) to a digital-signage product for behavior-driven content adaptation.

## Freelance

### Outlier AI

Remote

Mathematics & Language RLHF Expert

Jun 2024 – May 2025

- Domain expert evaluating and correcting large-language-model outputs for RLHF, on mathematics and regional-language reasoning projects.

## Leadership & Mentoring

### IIT Delhi

New Delhi, India

Research Mentor / Team Lead, DSIR Lab

2025 – Present

- Lead and mentor a team of B.Tech students and research interns in the lab, guiding them on multimodal retrieval and computer-vision research, experiment design, and scientific writing.

### Ideathon, SSGEC Bhavnagar

Bhavnagar, India

Team Leader — 1st Place

Feb 2022

- Won 1st place for proposing a renewable energy-storage solution that harnesses kinetic energy from flowing river water and converts it to stored potential energy.

## Teaching Experience

### G.S. Sanyal School of Telecommunications, IIT Kharagpur

Kharagpur, India

Teaching Assistant — Communications Simulation Lab & Embedded Communications Lab

Jul 2024 – May 2025

- Mentored 20 students through simulation and embedded-systems lab projects, with hands-on demonstrations and feedback.

## Skills

Languages: Python

ML / Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers

Retrieval & IR: FAISS (KNN / HNSW / IVFPQ / O-IVFPQ), Apache Solr, BM25

Multimodal Models: CLIP, OpenCLIP, FLAVA, UniIR, BLIP-2, Qwen-VL, MiniLM

Tools & Libraries: Git, FastAPI, Flask, Streamlit, NumPy, pandas, Matplotlib, OpenCV, Linux

Spoken Languages: English (full professional), Hindi (native)